Yimin Gao

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EDUCATION

Ph.D., Electrical Engineering, University of Virginia 2023–2027 (Expected)

M.S., Electrical Engineering, University of Virginia 2021–2022

B.E., Electrical Engineering, Univ. of Cincinnati & Chongqing Univ., Dual Degree Program 2015–2020

SUMMARY

Ph.D. student in computer architecture working on AI accelerators and HW/SW co-design for LLM inference. **First-author MICRO 2026 on LLM inference accelerators**, **two 2026 GF12 tapeouts in progress**. Full-stack from RTL and physical design through MLIR/IREE compiler backend and runtime.

PROJECTS

*FlexPosit: Tunable Fractional Precision for **LLM Inference** Accelerators* **MICRO 2026; GF12**

- Datapath: Designed a tunable fractional-precision posit datapath enabling per-kernel precision switching within a single bit-serial systolic array.
- **Quantization:** Built a mixed-precision post-training quantization pipeline in PyTorch matched to the hardware's supported numerics, reaching near-FP16 accuracy at sub-5-bit across LLaMA, OPT, Phi, and Yi.
- RTL + implementation: Implemented and verified the accelerator in SystemVerilog; validated end-to-end on FPGA and synthesized in a GlobalFoundries 12nm flow, now targeting a Dec. 2026 tapeout shuttle.
- Evaluation: Built a cycle-accurate simulator with memory and energy models (CACTI, Ramulator); measured **1.8× throughput** and **2.0× energy reduction** over prior LLM accelerators (OliVe, ANT).

*FlexNPU: Full-Stack FPGA NPU Platform **(Compiler, Runtime)***

- Full-stack bring-up: Built an end-to-end AI accelerator on a Xilinx Kria KV260 FPGA (Zynq SoC): custom compiler backend, C/C++ ARM-host runtime, and a double-buffered RTL shell with AXI DMA.
- **Compiler:** Wrote custom MLIR/IREE passes lowering ML ops onto the accelerator, plus a register/DMA interface allowing RTL datapath variants to be swapped without rebuilding the compiler.
- Performance analysis: Measured on-board GEMM against Timeloop predictions, attributing most wall time to overhead invisible to compute-cycle models (DMA, descriptor setup, control FSM).

*LiteAIR5: Mixed-Precision AI Acceleration on a Custom **RISC-V** SoC* **IEEE SOCC 2023**

- Re-architected a 16-bit integer multiplier into a SIMD dot-product unit with selectable precisions, where lower precision yields proportionally higher throughput, integrated via RISC-V ISA and microarchitecture co-design.
- Integrated the extended core into a full FPGA-based System-on-Chip; delivered **15.5–47.7×** speedup and up to **20.5 TOPS/W** on quantized CNN kernels over the baseline RISC-V core.

EXPERIENCE

Research Assistant, High-Performance Low-Power (HPLP) Lab, UVA 2022–Present

- Hand-drew a balanced CMOS PUF macro in **GF12**; LVS/DRC-clean, targeting the Dec. 2026 shuttle.
- Teaching Assistant for Computer Architecture & Design, Embedded Systems, and AI Hardware courses.

SKILLS

RTL/Verification: SystemVerilog, Verilog, Vivado, ModelSim
Physical Design: Synopsys DC/ICC2, Cadence Virtuoso (ADE, Spectre), LVS/DRC signoff, GF12
Programming: C/C++, Python, RISC-V Assembly, CUDA, Tcl, Bash, Git, Make
ML/Compilers/Modeling: PyTorch, MLIR, IREE, quantization, posit arithmetic, CACTI, Ramulator

SELECTED PUBLICATIONS

- **Y. Gao**, et al., "FlexPosit: Tunable Fractional Precision for LLM Inference Accelerators," **MICRO 2026, accepted**.
- X. Zhao, R. Xu, **Y. Gao**, et al., "Edge-MPQ: Layer-Wise Mixed-Precision Quantization," **IEEE Transactions on Computers**, 2024.
- **Y. Gao**, et al., "LiteAIR5: A system-level framework for the design and modeling of AI-extended RISC-V cores", **IEEE SOCC**, 2023.
- U.S. Patent Application: Membrane DRAM-PIM Filtering Architecture (Contributor), 2025.